Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (2)
MOSFET IV Characteristics

Spring 2023

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MOSFET IV Characteristics

Objectives

- To study the IV curve of a MOSFET.
- To graphically find the operating point of a MOSFET circuit.
- To experimentally evaluate the threshold voltage of a MOSFET.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in saturation and find expressions for the drain current and the drain-source voltage. Ignore channel length modulation.

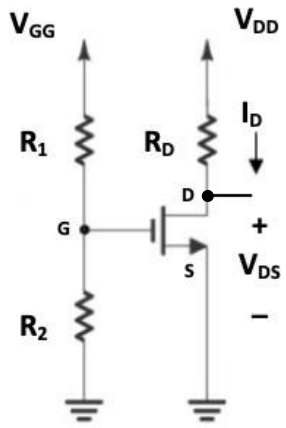


Figure 1: A MOSFET DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 6.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 1 kΩ (1), 1 MΩ (2)			
3	NMOS transistor (1)	2N7000	https://www.onsemi.com/pub/Collateral/2N7000-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): MOSFET IV Characteristics.

1. Connect the circuit in figure 1 with R_D=1 kΩ and R_1= R_2=1 MΩ. Pay a special attention to how you connect the terminals of the MOSFET (refer to the data sheet to identify the transistor terminals).

2. Set V_{GG} to a constant voltage of +8 V while connecting V_{DD} to a variable DC voltage source.
3. The IV characteristics (I_D versus V_{DS}) is obtained by slowly varying V_{DD} (from 0 to +10 V) and measuring the voltage V_{DS} and V_{R_D} at each setting. Use table 2 to enter your results.
4. Repeat step 3 for another value of V_{GG}, say +10 V. Record your results in table 2.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. Reconsider the circuit in figure 1, with $V_{GG} = V_{DD} = +10\,V$. Measure V_{DS} and record the results in table 3. Note that $I_D = (V_{DD} - V_{DS})/R_D$. This is the operating point of this circuit.

Part (3): MOSFET Threshold Voltage.

Produce a procedure to find the threshold voltage of the transistor using the circuit in figure 1. Write the steps below and implement them.

1. V_s is zero. Because V_s is connected to ground.
2. Therefore V_{Gs} is directly equal to the V_G.
3. We will use voltage divider to find V_G.
4. $10x\frac{1}{2} = 5V = V_G$.

For $V_{GG} = 8V$;
$V_{GS} = 8x\frac{1}{2} = 4V$

For $V_{GG} = 10V$;
$V_{GS} = 10x\frac{1}{2} = 5V$

Results and Analysis

Part (1): MOSFET IV Characteristics.

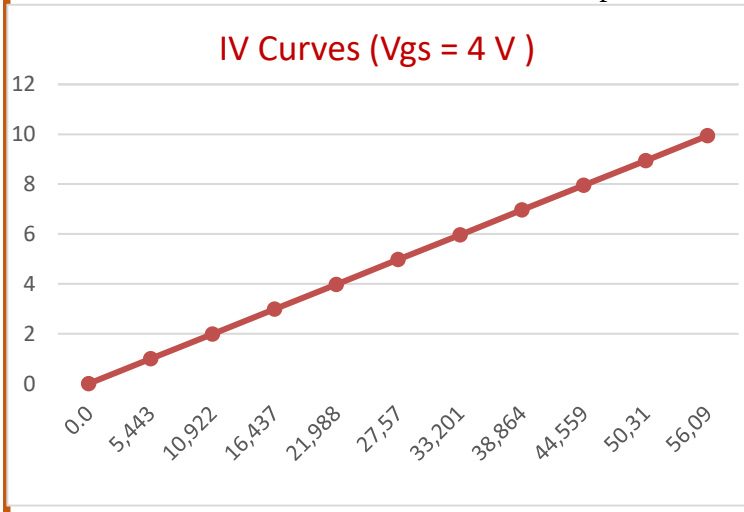
1. Calculate V_{GS} (using voltage divider rule) and the current I_D (V_{R_D}/R_D) in table 2, for both values of V_{GG}.
2. Use graph 1 to sketch the IV curves (I_D versus V_{DS}) for the two V_{GS} values using the results in table 2.
3. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

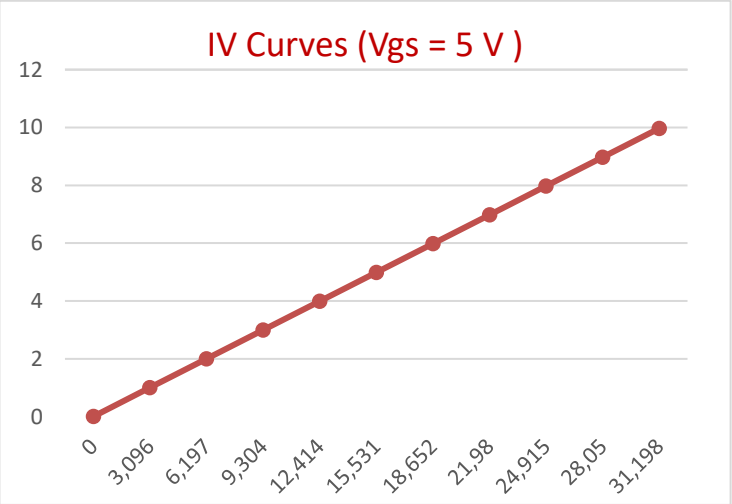
Answer:

Table 2. IV Characteristics Measurements.

	V_{GG} = 8V || V_{GS} = 4V			V_{GG} = 10V || V_{GS} =5V		
V_{DD} (V)	V_{DS}	V_{R_D}	I_D	V_{DS}	V_{R_D}	I_D
0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	5.443mV	994.6 mV	994.6 µA	3.096 mV	996.8 mV	996.8 µA
2.0	10.922mV	1.988 V	1.988 mA	6.197 mV	1.993 V	1.993 mA
3.0	16.437mV	2.983 V	2.983 mA	9.304 mV	2.990 V	2.990 mA
4.0	21.988mV	3.977 V	3.977 mA	12.414 mV	3.987 V	3.987 mA
5.0	27.570mV	4.972 V	4.972 mA	15.531 mV	4.984 V	4.984 mA
6.0	33.201mV	5.967 V	5.967 mA	18.652 mV	5.981 V	5.981 mA
7.0	38.864mV	6.961 V	6.961 mA	21.980 mV	6.978 V	6.978 mA
8.0	44.559mV	7.954 V	7.954 mA	24.915 mV	7.975 V	7.975 mA
9.0	50.31mV	8.948 V	8.948 mA	28.050 mV	8.972 V	8.972 mA
10.0	56.09mV	9.942 V	9.942 mA	31.198 mV	9.969 V	9.969 mA

Graph 1: MOSFET IV Curves.





4. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer:

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. The load line equation for this circuit is $I_D = \frac{VDD-VDS}{RD}$. It is the line connecting the two points (0,$\frac{V_{DD}}{R}$) and (V_{DD},0) on the IV curve.
2. Draw the load line on top of the IV curves in graph 1. Mark the intersection point of the curves.
3. The intersection of the load line and the IV curve gives the operating point of the circuit. Read the drain current (I_D) value and the drain-source voltage (V_{DS}) value from the intersection point in graph 1. Record the results in table 3.
4. The experimental value of $I_D = (V_{DD} - V_{DS})/R_D$.

$$I_d = (10 - 31.197x10^{-3})/1x10^3$$

$$I_d = 0.009968803\ A = 9.968803\ \text{mA}$$

$$\frac{10 - V_{DS}}{10^3} = 9.96880x10^{-3}$$

$$V_{DS} = 31.200\ \text{mV}$$

Table 3. Graphical Solution Results.

V_{DD}= 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference
31.197 mV	31.200mV	-0.0096%	9.969mA	9.968803 mA	0.002%

5. Comment on your results explaining the sources of differences.

Answers: The sources of differences between experimental and graphical values can vary depending on the experiment. In our situation we can say that Measurement Erors, Instrument Limitations.

Part (3): MOSFET Threshold Voltage.

1. Comment on your result regarding the threshold voltage as compared to the data sheet value.
2. Comment on the procedure that you used to find the threshold voltage. How can it be improved?
3. Can you estimate the value of the constant k_n for this transistor? Show your work.

Answers: We calculated 5 volts on our mosfet. We checked the data sheet of the mosfet. The maximum threshold voltage of the mosfet is 3 in our mosfet. These differences may be due to excessive input voltage, incorrect calculations and measurement techniques.

Conclusions

In this experiment, we analyzed a MOSFET type transistor's behavior in which the source part is connected to ground.

The drain part is connected to a DC source (Vdd) and a 1kΩ resistor (Rd). The gate part is connected to another DC voltage value and placed between 2 resistors. We observed that as we increase the Vdd voltage; Vds, Vrd and Id is also increases proportionally.

We also observed that the gate voltage is always half of the Vgg voltage because resistors are used as a voltage divider and both resistors have the same resistance. For example, we first assign Vgg as 8V and the gate voltage becomes 4V. Then, we assigned Vgg as 10V and the gate voltage becomes 5V. On the other hand, despite the gate has a voltage,there is no current through the gate.

Name	ID	Section
Yunus Emre Turunç	150721854	Online

2N7000G

Small Signal MOSFET 200 mAmps, 60 Volts

N–Channel TO–92

Features

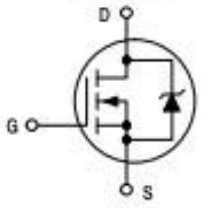
- AEC Qualified
- PPAP Capable
- This is a Pb–Free Device*

ON Semiconductor®

http://onsemi.com

200 mAMPS
60 VOLTS
$R_{DS(on)}$ = 5 Ω

N–Channel



MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain Source Voltage	V_{DSS}	60	Vdc
Drain–Gate Voltage (R_{GS} = 1.0 MΩ)	V_{DGR}	60	Vdc
Gate–Source Voltage – Continuous – Non–repetitive ($t_p \le 50$ μs)	V_{GS} V_{GSM}	±20 ±40	Vdc Vpk
Drain Current – Continuous – Pulsed	I_D I_{DM}	200 500	mAdc
Total Power Dissipation @ T_C = 25°C Derate above 25°C	P_D	350 2.8	mW mW/°C
Operating and Storage Temperature Range	T_J, T_{stg}	–55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction–to–Ambient	$R_{\theta JA}$	357	°C/W
Maximum Lead Temperature for Soldering Purposes, 1/16" from case for 10 seconds	T_L	300	°C

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

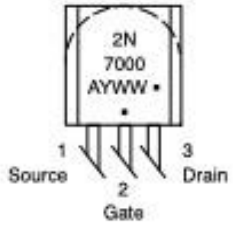
TO–92
CASE 29
STYLE 22



STRAIGHT LEAD
BULK PACK

BENT LEAD
TAPE & REEL
AMMO PACK

MARKING DIAGRAM AND PIN ASSIGNMENT



A = Assembly Location
Y = Year
WW = Work Week
▪ = Pb–Free Package
(Note: Microdot may be in either location)